

Supplementary Table 11. Transformer input, tokenization, masking, objective, optimizer and training hyperparameters

Frontiers in Signal Processing R1 supplementary table

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Standalone display version for LaTeX upload/review. Machine-readable source: `source_tables/Supplementary_Table_11_transformer_hyperparameters.csv`. Rows: 28; columns: 5. The display below contains the complete source table.

Table 1: Supplementary Table 11

section	parameter	selected value	sweep or fixed values	notes
Input	signal channels	FHR and UC	FHR and UC	Two-channel CTG windows were used for SSL pretraining.
Input	window length	10 minutes	Fixed	Each window contained 2400 samples per channel at 4 Hz.
Input	sampling rate	4 Hz	Fixed	Signals were resampled or represented at the project target sampling rate.
Tokenization	patch length	80 samples	80, 120	Non-overlapping patches were used in the selected model.
Tokenization	patch stride	80 samples	Equal to patch length	The selected configuration produced 30 temporal patches per channel.
Tokenization	token count	60 channel-time tokens	Derived	Thirty temporal patches by two channels.
Encoder	model dimension	64	48, 64	Latent token dimension before transformer encoding.
Encoder	transformer layers	3	2, 3	Compact encoder depth for the selected manuscript-candidate.
Encoder	attention heads	4	Fixed	Multi-head self-attention heads.
Encoder	feedforward width	4 x model dimension	Fixed	Transformer feed-forward layer width.
Encoder	activation	GELU	Fixed	Activation function in transformer feed-forward layers.
Encoder	dropout	0.1	Fixed	Applied within the compact transformer encoder.
Masking	masking strategy	Channel masking	Mixed, block, channel	Channel masking was used by the held-out manuscript-candidate; mixed and block masking were sensitivity configurations.
Masking	mask fraction	0.25	Fixed	Fraction of patches or channel-time tokens masked in each view.
Augmentation	jitter standard deviation	0.02	Fixed	Gaussian jitter applied to paired training views.
Objective	reconstruction loss	Masked-patch mean squared error	Fixed	Computed only over masked patches.
Objective	contrastive loss	NT-Xent	Fixed	Applied to projected pooled embeddings from paired views.
Objective	contrastive temperature	0.2	Fixed	Temperature parameter for NT-Xent loss.
Objective	contrastive weight	0.2	0.0, 0.2	Weight applied to the contrastive objective in the total loss.
Projection	projection dimension	64	Fixed	Projection head output dimension for contrastive learning.
Training	optimizer	AdamW	Fixed	Optimizer used for compact candidate training.
Training	learning rate	0.001	Fixed	Learning rate for AdamW.
Training	weight decay	0.0001	Fixed	Weight decay for AdamW.
Training	batch size	32	Fixed	Mini-batch size for SSL training.

Continued from Supplementary Table 11.

section	parameter	selected value	sweep or fixed values	notes
Training	gradient norm clipping	1	Fixed	Maximum gradient norm.
Training	candidate epochs	4	Fixed	Candidate configurations in the compact sweep were trained for four epochs.
Training	selected architecture trace	12 epochs	Inspection only	The 12-epoch trace was used for training-dynamics inspection and did not determine model selection.
Downstream	record embedding	Mean pooled window embeddings	Fixed	Clean unmasked windows were encoded and averaged to record-level SSL representations.